Monica Luong

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Education

Case Western Reserve University - School of Engineering Cleveland, OH

Bachelor of Science in Computer Science, ***GPA: 3.9*** *(Expected) May 2028*

- **Coursework**: Programming in Java, Data Structures, Logic Design and Computer Organization
- **Activities**: Grace Hopper, Women in Tech, Klover, Rewriting the Code, CodePath, Girls Who Code

Experience

Undergraduate Research Assistant January 2026 – Present

Case Western Reserve University *Cleveland, OH*

- Leading research efforts to evaluate and improve the safety of **AI coding agents** by analyzing model behavior on real-world software development workflows.
- Engineering a scalable data pipeline to ingest and analyze **1,000+ GitHub issues**, benchmarking the performance of multiple AI models on complex technical issue classification.
- Increased model accuracy by **18%** and reduced inference overhead by **25%** by iteratively refining prompt structure, quantifying trade-offs between prompt complexity, latency, and computational cost.
- Developed an AI safety evaluation framework connecting PR-level AIDev workflow data with coding-agent failure cases, reducing analysis time by **30%** across accountability, privacy, security, and code quality risks.
- Implemented a reproducible analysis pipeline in **Python** using **StarCoder2** and **LLaMA3** to extract PR metadata, disclosure markers, and review outcomes, scaled on a **40-GPU H200 HPC cluster** for large-scale inference.

Technical Office Assistant September 2025 – Present

Case Western Reserve University *Cleveland, OH*

- Streamlined daily operations for **25+** students, faculty, and staff by standardizing request tracking, resource organization, and checklist-driven workflows to improve reliability and follow-through.
- Applying systematic task tracking and prioritization practices to manage concurrent requests, reducing missed tasks and improving response time for departmental needs.

Projects

AGEent (Amazon Nova AI Hackathon) | *Next.js, FastAPI, PostgreSQL, Amazon Bedrock, Amazon Nova*

- Developed an **AI-powered voice assistant** that guides seniors through complex government processes (Medicare, Social Security, DMV), replacing fragmented multi-portal workflows with a single conversational interface and reducing task completion time by **65%**.
- Automated end-to-end government form completion via voice, enabling seniors to complete tasks hands-free and cutting average workflow duration from **30 minutes to 10 minutes**.
- Integrated **Amazon Nova Act** for autonomous form navigation and completion, **Nova Sonic** for real-time voice-to-action input, and **Nova Lite** for multi-step reasoning and task orchestration.

Outfits Recommendation System | *Python, React, Vercel, OpenSource services*

- Built a full-stack application that automated daily outfit recommendations for **100+ active users** by integrating real-time weather APIs.
- Designed a machine learning recommendation system that transformed weather features into personalized outfit suggestions, reducing users' daily decision time by **60%**.
- Delivered features iteratively using Agile workflows, improving development velocity and code reliability.

SpotOn | *React, Leaflet, Vite, LocalStorage*

- Led a team of 3 to build a responsive web application used by **5,000+ students** to discover real-time campus study spaces, reducing average search time for open locations by **40%** during peak hours.
- Implemented an interactive Leaflet map displaying 14+ high-traffic campus locations with dynamic, color-coded availability indicators, improving space discovery accuracy and user engagement by **30%**.
- Optimized the frontend for mobile-first performance using Vite and LocalStorage caching, cutting initial load times by **35%** and enabling seamless, low-latency usage for on-the-go students.

Technical Skills

Languages: Java, Python, HTML/CSS, JavaScript, TypeScript
Frameworks & Libraries: React, Next.js, FastAPI, NumPy, Pandas, Matplotlib, scikit-learn
Technologies: Git, Github, Vercel, Docker, MacOS, Linux, VS Code, PostgreSQL, Google Cloud, AWS
Interests: Full Stack Development, Artificial Intelligent, Machine Learning